

Sustainable Sites Initiative

The Sustainable Sites Initiative takes green building standards outside.

by **Heather Venhaus**

It's a valuable addition to a home or business—well-landscaped grounds with eye-pleasing greenery and colorful flowers. How good it would be if those attractive landscapes were also saving energy and contributing to sustainability!

Increasingly, designers, planners, builders, and developers are looking at the landscapes that surround the homes they build and responding to client requests for energy conservation outside as well as in. And many cities recognize the critical role landscapes play in increasing property values and reducing the cost of providing water and handling storm water runoff.

The Sustainable Sites Initiative is taking green building standards outside. This partnership of the American Society of Landscape Architects, the Lady Bird Johnson Wildflower Center, and the United States Botanic Garden is working with a diverse group of supporting organizations to develop the first national guidelines and standards for sustainable landscapes. The result will be incentives to create large and small commercial and noncommercial landscapes that contribute to the good of the environment and improve our overall quality of life.

This effort grew from the recognition on the part of landscaping professionals and others that there are no national green standards for landscapes as there are for buildings. The United States Green Building Council (USGBC) and other organizations have established national green building rating systems, such as LEED (Leadership in Energy and

Environmental Design). But so far, only local guidelines have existed for green landscapes.

The Sustainable Sites Initiative brings together a number of programs and organizations. These include USGBC, EPA's GreenScapes program, the National Recreation and Parks Association, the American Society of Civil Engineers' Environment and Water Resources Institute, the National Association of County and City Health Officials, the Nature Conservancy's Global Invasive Species Initiative, and the Center for Sustainable Development at the University of Texas at Austin. Representatives of these organizations guide and direct the development of the green landscapes rating system.

The Product Development Committee is assisted by 32 experts in such fields as landscape architecture, horticulture, conservation, sustainable

design, civil and environmental engineering, ecology, hydrology, forestry, soils, planning, public health, outdoor recreation, and other disciplines that influence the design, construction and maintenance of landscapes.

The Sustainable Sites Initiative has released a preliminary report, available at www.sustainablesites.org, with 200 recommendations that can benefit everyone in the site development industry. Already, more than 450 industry professionals have responded to our request for feedback. We will use this feedback to prepare the final standards and guidelines, to be published in 2009. The U.S. Green Building Council, a stakeholder in the initiative, anticipates incorporating these standards and guidelines into future revisions of the LEED Green Building rating system.

Ideally, landscaping is of enormous benefit to the environment. Soil



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This inviting front yard of wine cups uses native plants as one element of a sustainable landscape.



SALLY AND ANDY WASOWSKI

Cooperation with native plants creates a backyard oasis.

supports plant growth and absorbs water, reducing runoff and erosion, and even filtering chemical pollutants out of water. Even more exciting is the way that plants and soils can capture carbon dioxide (CO₂)—an important greenhouse gas—hold it in the plant tissues, and deposit it in the soil, preventing it from entering the atmosphere. With all of the bad news about global warming and the complicated efforts to prevent more damage, this elegant partnership with nature is refreshingly simple. In addition, strategically placing vegetation to shade a building from the sun and serve as a windbreak can reduce that building's energy consumption by up to 25%.

Not only do plants take CO₂ from the air, they also filter out other pollutants, such as ozone, particulates, sulfur dioxide (SO₂), and carbon monoxide (CO). A 2007 study of New York City's urban forest estimated that each tree intercepted 1.73 lb of air pollutants per year. These citywide trees intercepted 129.1 tons of ozone and 63 tons of particulate matter per year. This study estimated that New York City trees provided more than \$100 million in net annual benefits, including electricity savings, CO₂ sequestration, air pollutant removal, storm water runoff reduction, and increased property values. This averaged out at \$5.60 per tree in benefits, compared to \$1 per tree spent on maintenance.

The net result of a well-designed, carefully constructed landscape is energy conservation for both the landscape and the home. That means dollars in the homeowner's pocket, as well as the satisfaction of doing good for the environment.

The Sustainable Sites Initiative's primary recommendation is to identify and preserve the elements of a site that contribute to energy savings and benefit the environment. The process starts with a site assessment that will guide the design and construction of a building, and even the maintenance, after the building project is completed.

The preliminary report described above contains more than 200 strategies developed from best practices of industry professionals, as well as



PHILIP HAWKINS

A green roof on a shopping center in Austin, Texas, is useful in reducing the energy needs of the building as well as reducing stormwater runoff from the roof. Plantings include rock roses and Mexican feather grass.

research on techniques that cooperate with the givens of a building site to save energy and minimize impact on the site and the environment. Let's talk about just a few of these strategies for designing and maintaining sustainable landscapes.

Maintain or improve the health of the soil. Healthy topsoil—which varies in depth depending on location—is enormously beneficial. It minimizes runoff, stores water, and provides a healthy rooting environment and habitat to a wide range of useful organisms. A single grain of soil can contain from 1,000 to 10,000 different species of bacteria and fungi. One very useful service performed by bacteria is nitrogen fixation, the process by which nitrogen is taken from its natural state in the atmosphere and converted into nutrients for plants.

Therefore, it's best to disturb healthy soil as little as possible. Heavy equipment on wet soil, roadways, and construction staging areas causes compaction that limits water infiltration and plant growth. Confine staging, utility lines, and roads to areas where the soil has already been disturbed. Then work closely with your contractors to restrict equipment traffic to those designated areas. Where traffic is unavoidable, try to protect against rutting and compaction.

To protect healthy soil, remove topsoil from the building site and store it for reuse in landscaping. Not only does that save trucking in more topsoil, but it retains soil uniquely adapted to the location.

Avoid the use of chemicals that harm the health of all organisms.

The U.S. Fish and Wildlife Service reports that homeowners use up to 10



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Groundcover such as Virginia Creeper can create a landscape that uses less water than a conventional lawn.

times more chemical pesticides per acre on their lawns than farmers use on crops. These pesticides and other chemicals used in lawn care can be carried into streams, rivers, and lakes. EPA reports that urban runoff is one of the leading sources of pollutants that affect more than 260,000 miles of streams and rivers.

That's one reason to minimize the use of chemical fertilizers. When you are ready to plant, test the soil for nutrients—healthy soil may not need much help. If you need to add nutrients, it's best to use organic matter, such as compost. You can make compost yourself by collecting a pile of plant matter—leaves, lawn clippings, vegetable parings—and soil, moistening it, and allowing it to decompose. Or you can purchase organic soil amendments.

Be cautious also in using pesticides and herbicides. Choose the least toxic alternative. Native plants often require very little assistance in handling pests. You can also learn about integrated pest management, an approach to pest control that relies on life cycles and biology rather than on chemicals.

Reduce waste. Recycling yard waste on-site can be easy, and it can make a big difference to soil and plant health. In the United States, leaves and grass make up as much as 18% of landfill, while another 7% is made up of rock, soil, and woody landscape waste. In Texas alone, it costs \$150 million a

year to dispose of this yard waste (Texas Commission on Environmental Quality, 2005). Don't pay to throw this waste away; use plant material as organic fertilizer after you compost it. Rocks and lumber can be used to help reshape your landscape. Or just leave rock formations where they are—it saves labor—rather than fighting Mother Nature.

Water is a scarce resource. Value it! Think of this: The turf grasses used in residential and commercial lawns and golf courses are the single largest irrigated crop in America, covering an area the size of the state of Mississippi. Nationwide, landscape irrigation is estimated by EPA to account for almost one-third of all residential water use, totaling more than 7-billion gallons per day. You can reduce the need for water and save money, by capturing and using rainwater. Design rain gardens to use water that might otherwise run off the site. Rain gardens use water-tolerant plants to slow runoff (and create eye appeal in what might otherwise be unattractive drainage). Mulching your plants also conserves water, as does grading designed to capture and hold rainwater. Finally, remember that plants adapted to the region are less apt to be water hogs.

Select your landscape materials wisely. More and more, we need to think about the energy embodied in

the materials we use to enhance the outdoors, whether they are paving blocks, decks, fences, or other hardscape. It takes energy to manufacture and transport these materials. Think local and try to find materials that do not require expensive shipping. Consider reusing or recycling the materials in the fence or deck you are replacing. Don't order more materials than you need, and try to minimize the waste you ship away—shipping uses energy, too.

Durable materials last longer, saving the energy and money that it would otherwise cost to replace them. Specify recycled materials or sustainably harvested forest products for your landscape. Locally grown plants not only save resources, but are better adapted to your location.

Reduce energy use in the landscape. Many cities and suburbs are warmer than the countryside that surrounds them because the concentration of pavement and buildings absorbs heat. Landscapes reduce the urban heat island effect by providing shade and releasing moisture. Trees can shade roofs, reducing indoor temperatures. Open-grid paving and pervious or semipervious pavement is cooled by water and air passing through it. All of these features reduce the demand for energy to cool nearby buildings.

More and more, we have learned to think about energy savings in constructing buildings. The Sustainable Sites Initiative is an effort to create landscapes that not only save energy, but save water, reduce air and water pollution, and benefit both the homeowner and the environment. **He**

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For more information:

To learn more about the Sustainable Sites Initiative, go to www.sustainable-sites.org.